Seminar Quiz 1

Empirical Analysis for Policymakers

Sciences Po Paris - Reims

25 minutes. No extra papers. You are free to use calculators.

MCQs

1. Which of the following is true about independent events?

   A) If two events are independent, Pr(A | B) = Pr(B)

   B) If two events are independent, Pr(A | B) = Pr(A)

   C) If two events are independent, Pr(A, B) = Pr(A) + Pr(B)

   D) If two events are independent, Pr(A, B) = Pr(A or B)

2. If the probability of drawing an ace from a deck of 52 cards is 0.077, what is the probability of drawing an ace without replacement after already drawing one?

   A) 0.059

   B) 0.078

   C) 0.039

   D) 0.067

3. In the OLS regression model, the goal is to minimize which of the following?

   A) Sum of squared residuals

   B) Sum of squared errors

   C) Sum of expected values

   D) Sum of probabilities

4. You have a dataframe `df` and you want to filter out all rows where the column `age` is greater than 18. What is the next step in the pipe (`%>%`) after starting with `df`?

   A) `arrange(desc(age))`

   B) `filter(age > 18)`

   C) `select(age > 18)`

D) `group_by(age)`

5. The intercept (β_0 or α) in a simple linear regression represents:

   A) The change in Y for a unit change in X

   B) The expected value of Y when X = 0

   C) The residual error

   D) The slope of the regression line

6. What is the joint probability of two independent events A and B, if Pr(A) = 0.5 and Pr(B) = 0.3?

   A) 0.8

   B) 0.15

   C) 0.2

   D) 0.5

7. Which of the following is not a property of the expected value?

   A) $E(c) = c$, for any constant c

   B) $E(aX + b) = aE(X) + b$, for constants a, b

   C) $E(aX_1 + bX_2) = aE(X) + bE(X_2)$, for constants a, b

   D) $E(2X) = 4E(X)$

True/False Questions

1. The sum of the probabilities of all outcomes in a sample space is always 1.

2. In R, the `filter()` function from the `dplyr` package allows you to select specific columns of a dataframe.

3. In a simple linear regression model, the residuals and the error term are the same.

4. In R, `mutate()` can be used to create new variables in a dataframe.

5. Variance measures the spread or dispersion of a random variable around its expected value.

Short Essay Question

You are interested in studying the effect of legal drinking ages across US states on the rates of drinking and driving incidents. Explain the steps you would take to set up the model using OLS regression, and discuss any assumptions you would make about the error term and the relationship between the variables. (2 points + 1 bonus for showing it on the regression formula)

Mathematical Questions

1. **Bayes' Rule Problem:** A factory has two machines. Machine A produces 70% of the items, and Machine B produces 30%. 2% of the items from Machine A are defective, while 6% of the items from Machine B are defective. If an item is chosen at random and is found to be defective, what is the probability it was produced by Machine A? (2 points)

   Recall the Bayes' Rule in this context:

   $$\Pr(A|D) = \frac{\Pr(D|A) \cdot \Pr(A)}{\Pr(D)}$$

2. **Expected Value and Variance:** A government is analyzing the average weekly income of individuals in a certain region. Let X represent the weekly income (in dollars). The population mean of weekly income is known to be $\mu = 800$, and the population variance is $\sigma^2 = 1900$. (4 points)

Recall the population and sample variance formulae:

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(X_i - \mu)^2 \quad \text{and} \quad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

a) What is the expected value $E(X)$ of the weekly income?

b) What is the population variance $Var(X)$?

c) Suppose you have a sample of weekly income data from 10 individuals: 780, 800, 820, 760, 840, 860, 880, 740, 800, 820. Calculate the sample variance s^2 of these incomes.

d) Comparing the sample and the population variances, would you say that the sample is random enough to represent the population and why?